

# Lab 2: Hazard-Damage Convolution

CEVE 421/521

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## 1 Overview

In lecture, we learned that **risk** emerges from the interaction of hazard, exposure, and vulnerability. Today, we'll implement the mathematical machinery that connects them: **convolution**.

Convolution is how we transform a distribution of hazards into a distribution of damages. This is the core computational technique behind flood risk scores, insurance pricing, and cost-benefit analysis.

### Learning Objectives:

1. Understand convolution as the bridge between hazard and damage distributions
2. Implement convolution both analytically (simple case) and via Monte Carlo
3. Work with the GEV distribution for extreme events
4. Apply a depth-damage function to compute flood losses
5. Extend analysis to a community of 100 houses
6. Get introduced to the SimOptDecisions.jl framework

## 2 Before Lab

Complete these steps **before** coming to lab. Read through the entire notebook so you understand what we'll be doing.

### 2.1 Setup

Julia uses packages to make code reusable and modular. Run this block to install all required packages – the first time may be slow while everything is installed and compiled.

```
using Pkg
try
    # Activate the local project
    Pkg.activate(@__DIR__)

    # Check if Manifest exists, if not instantiate
    if !isfile(joinpath(@__DIR__, "Manifest.toml"))
        Pkg.instantiate()
    else
        # Try to resolve any inconsistencies
        Pkg.resolve()
    end
end
```

```

catch e
  @warn "Package environment setup failed, attempting recovery" exception=e
  Pkg.activate(@__DIR__)
  Pkg.instantiate()
end

```

Now load the packages we'll use:

```

using CairoMakie
using Distributions
using Random
using Statistics

Random.seed!(2026)

```

#### Line 1

For creating figures

#### Line 2

For probability distributions (GEV, Uniform, etc.)

#### Line 3

For reproducible random numbers

#### Line 4

For `mean()` and other statistics

#### Line 6

Set seed so everyone gets the same results

## 2.2 Convolution by Hand

Before we use computers, let's understand convolution with a simple example you can do with pencil and paper.

### 2.2.1 A Simple Flood Risk Problem

Imagine a house worth \$100,000 that faces flood risk. Based on historical data, we know:

| Event          | Probability | Damage   |
|----------------|-------------|----------|
| No flood       | 87%         | \$0      |
| Minor flood    | 10%         | \$10,000 |
| Moderate flood | 2%          | \$50,000 |
| Major flood    | 1%          | \$90,000 |

### 2.2.2 Computing Expected Damage

The **expected annual damage (EAD)** is the probability-weighted average of all possible damages:

$$\text{EAD} = \sum_i p_i \times D_i$$

### Your Response

*Compute the expected annual damage by hand. Show your calculation.*

*Replace this text with your work.*

Let's verify with code:

```
probabilities = [0.87, 0.10, 0.02, 0.01]
damages = [0, 10_000, 50_000, 90_000]
expected_damage = sum(probabilities .* damages)
println("Expected Annual Damage: \${$(expected_damage)}")
```

#### Line 1

Probabilities must sum to 1.0

#### Line 2

Damages in dollars for each scenario

#### Line 3

The .\* does element-wise multiplication; sum() adds them up

```
Expected Annual Damage: $2900.0
```

This is **convolution**: we've transformed a hazard distribution (flood probabilities) into a damage distribution and computed its expected value.

### Your Response

*A homeowner asks: "My expected damage is only \$2,900/year, so floods aren't a big deal, right?"*

*What would you tell them? Think about what expected value hides.*

*Replace this text with your answer.*

## 3 During Lab

Work through these sections during the lab session.

### 3.1 Monte Carlo Convolution

With continuous distributions and nonlinear damage functions, analytical convolution becomes intractable. Monte Carlo simulation offers a flexible alternative.

### 3.1.1 The GEV Distribution

The **Generalized Extreme Value (GEV)** distribution models annual maximum flood stages. It has three parameters:

- $\mu$  (**location**): roughly the “typical” annual maximum
- $\sigma$  (**scale**): how spread out the distribution is
- $\xi$  (**shape**): tail behavior;  $\xi > 0$  means fat tails (more extreme events)

```
 $\mu$  = 5.0  
 $\sigma$  = 2.0  
 $\xi$  = 0.1  
  
surge_dist = GeneralizedExtremeValue( $\mu$ ,  $\sigma$ ,  $\xi$ )
```

#### Line 1

Location parameter (ft) – shifts the distribution left/right

#### Line 2

Scale parameter (ft) – must be positive

#### Line 3

Shape parameter – positive means heavy right tail

#### Line 5

Create a GEV distribution object from the Distributions.jl package

Let's visualize this distribution:

```
let  
    x_range = range(0, 15; length=200)  
    fig = Figure()  
    ax = Axis(  
        fig[1, 1];  
        xlabel="Flood Stage (ft)",  
        ylabel="Density",  
        title="Annual Maximum Flood Stage Distribution"  
    )  
    lines!(ax, x_range, pdf.(surge_dist, x_range); linewidth=2)  
    fig  
end
```

#### Line 2

Create 200 evenly-spaced points from 0 to 15

#### Line 10

`pdf.()` broadcasts the probability density function over all x values

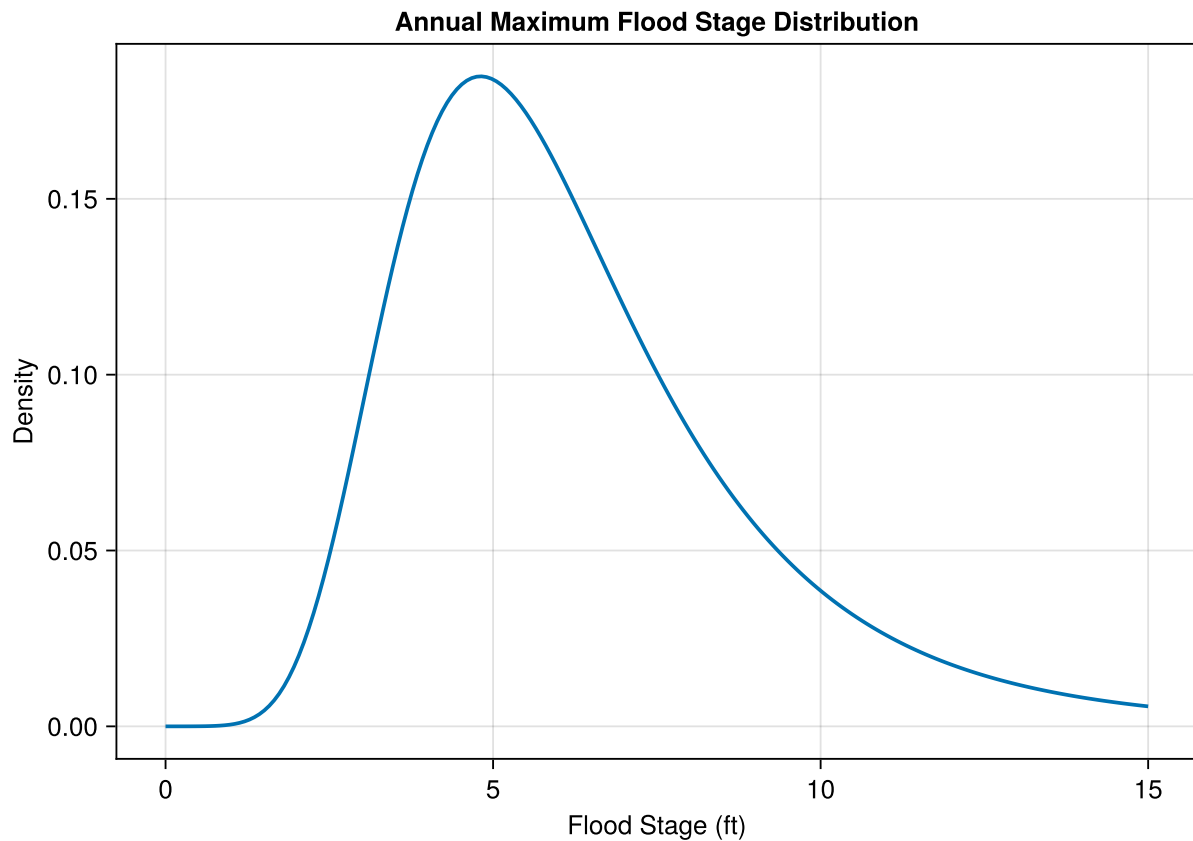


Figure 1: GEV distribution for annual maximum flood stage.

### 3.1.2 Exploring the GEV Parameters

Try changing the GEV parameters below to build intuition. Watch how the distribution shape changes.

```
let
  x_range = range(0, 20; length=200)
  fig = Figure(size=(700, 300))

  # Try changing these values!
  μ_test = 5.0 # Try: 3.0, 5.0, 7.0
  σ_test = 2.0 # Try: 1.0, 2.0, 4.0
  ξ_test = 0.1 # Try: -0.1, 0.0, 0.1, 0.3

  test_dist = GeneralizedExtremeValue(μ_test, σ_test, ξ_test)

  ax = Axis(
    fig[1, 1];
    xlabel="Flood Stage (ft)",
    ylabel="Density",
    title="GEV(μ=$μ_test, σ=$σ_test, ξ=$ξ_test)"
  )
```

```

lines!(ax, x_range, pdf.(test_dist, x_range); linewidth=2)
fig
end

```

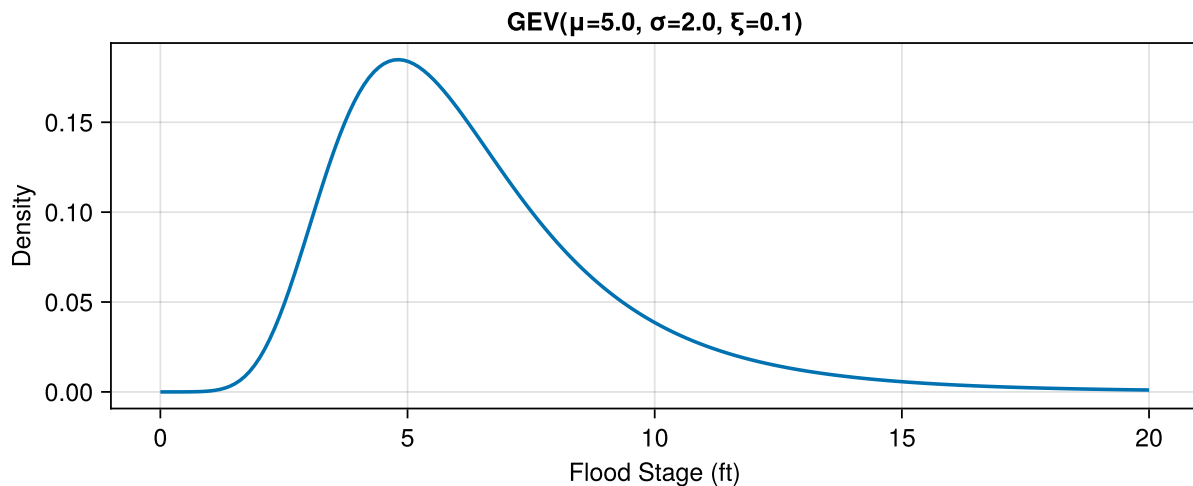


Figure 2: Experiment with different GEV parameters to see how they affect the distribution.

### 💡 Your Response

*Experiment with the parameters above. What happens when you:*

- Increase  $\mu$  from 5 to 7?
- Increase  $\sigma$  from 2 to 4?
- Change  $\xi$  from 0.1 to 0.3?

*Replace this text with your observations.*

### 3.1.3 Monte Carlo Sampling

We'll draw 10,000 samples from the GEV distribution. Each sample represents one possible year's annual maximum flood.

```

n_samples = 10_000
hazard_samples = rand(surge_dist, n_samples)

```

#### Line 2

`rand(distribution, n)` draws `n` random samples from the distribution

We plot the samples as a **normalized histogram** (probability density) so we can compare it to the true PDF. The area under the histogram equals 1.

```

let
    fig = Figure()

```

```

ax = Axis(
    fig[1, 1];
    xlabel="Flood Stage (ft)",
    ylabel="Density",
    title="10,000 Sampled Annual Maximum Flood Stages"
)
hist!(ax, hazard_samples; bins=50, normalization=:pdf)
lines!(ax, range(0, 15; length=200), pdf.(surge_dist, range(0, 15; length=200));
    color=:red, linewidth=2, label="True PDF")
axislegend(ax; position=:rt)
fig
end

```

#### Line 9

`normalization=:pdf` scales the histogram so the total area equals 1

#### Line 11

Overlay the true PDF to show our samples match the distribution

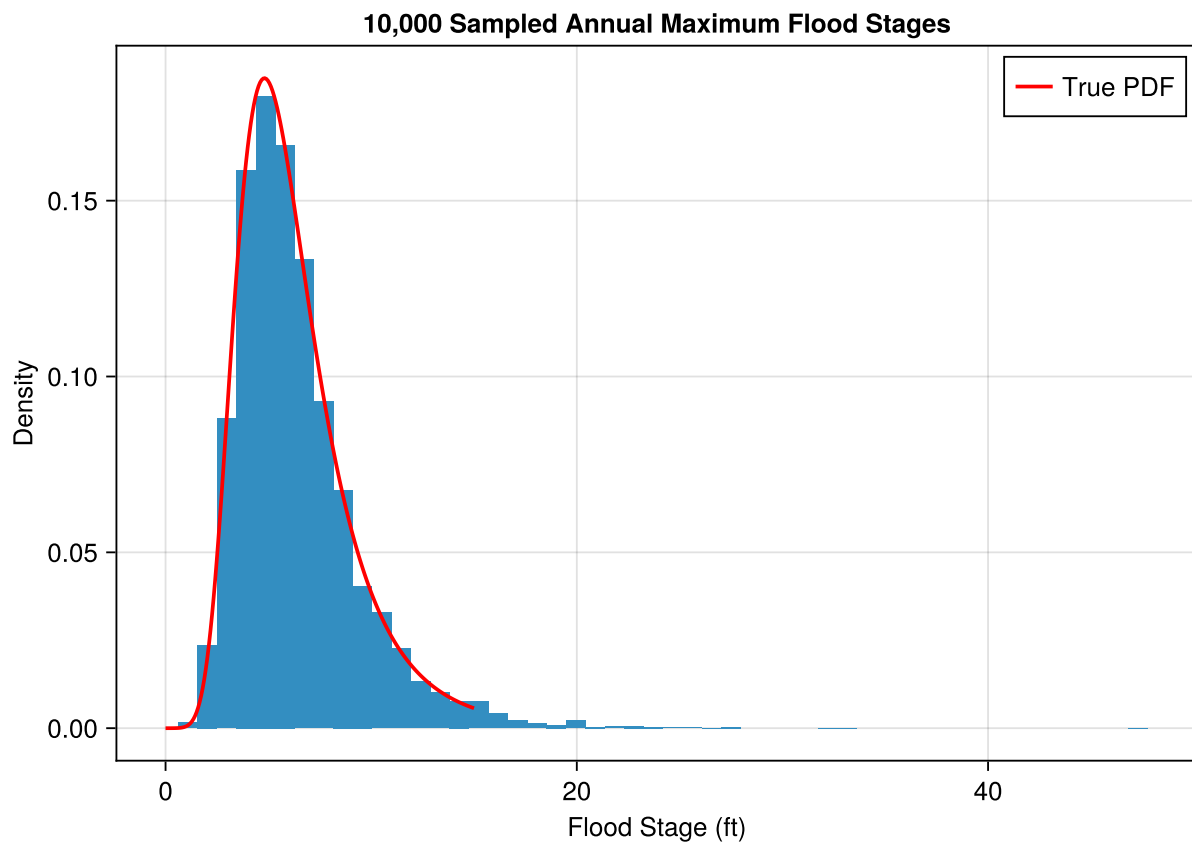


Figure 3: Monte Carlo samples from the flood stage distribution, normalized as a PDF.

### 3.1.4 A Depth-Damage Function

Now we need a function that converts flood depth to damage. We'll use a **logistic** (S-shaped) curve: damage starts slowly, increases rapidly in the middle range, then saturates.

```
"""Logistic damage function. Returns damage in dollars."""
function logistic_damage(
    flood_stage::T, house_elevation::T, house_value::T;
    threshold::T=one(T), saturation::T=T(7)
) where {T<:AbstractFloat}
    depth = max(zero(T), flood_stage - house_elevation)
    depth <= threshold && return zero(T)
    depth >= saturation && return house_value
    midpoint = (threshold + saturation) / 2
    steepness = T(6) / (saturation - threshold)
    damage_fraction = one(T) / (one(T) + exp(-steepness * (depth - midpoint)))
    return damage_fraction * house_value
end
```

#### Line 5

Type parameter ensures all inputs have consistent floating-point type

#### Line 6

Flood depth = water level minus floor elevation (can't be negative)

#### Line 7

No damage below 1 ft of water inside the house (furniture above floor, etc.)

#### Line 8

Total loss at 7 ft of water

#### Line 9

Logistic curve centered at midpoint between threshold and saturation

#### Line 11

Standard logistic function formula

Let's visualize this damage function:

```
let
    house_elev = 2.0
    house_val = 100_000.0
    flood_stages = range(0, 12; length=200)
    damages = [logistic_damage(s, house_elev, house_val) for s in flood_stages]

    fig = Figure()
    ax = Axis(
        fig[1, 1];
        xlabel="Flood Stage (ft)",
        ylabel="Damage (\$)",
        title="Damage Function (House at 2 ft elevation)"
    )
```



```

)
lines!(ax, flood_stages, damages; linewidth=2)
vlines!(ax, [house_elev]; color=:gray, linestyle=:dash, label="House floor")
axislegend(ax; position=:lt)
fig
end

```

#### Line 5

List comprehension: apply the function to each flood stage

#### Line 15

Vertical line showing where the house floor is

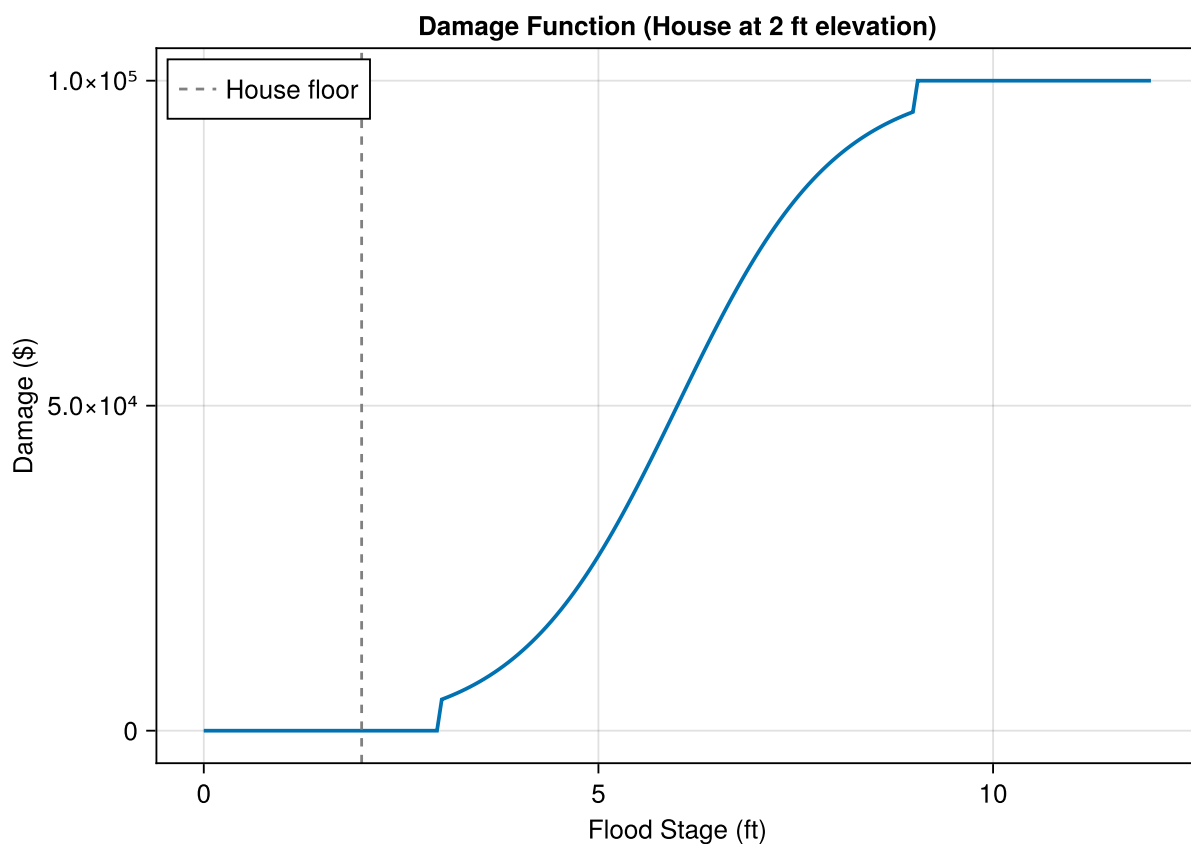


Figure 4: Logistic depth-damage function for a house at 2 ft elevation.

### 3.1.5 Convolution via Monte Carlo

Now we convolve the hazard samples through the damage function. This transforms our distribution of flood stages into a distribution of damages.

```

house_elevation = 2.0 # feet
house_value = 100_000.0 # dollars

```

```
damage_samples = [logistic_damage(h, house_elevation, house_value) for h in
hazard_samples]
```

#### Line 4

Apply the damage function to each of the 10,000 flood stage samples

Again, we plot as a normalized histogram (probability density). Notice the spike at \$0 (years with no damage) and the long right tail.

```
let
  fig = Figure()
  ax = Axis(
    fig[1, 1];
    xlabel="Annual Damage (\$)",
    ylabel="Density",
    title="Distribution of Annual Damages"
  )
  hist!(ax, damage_samples; bins=50, normalization=:pdf)
  fig
end
```

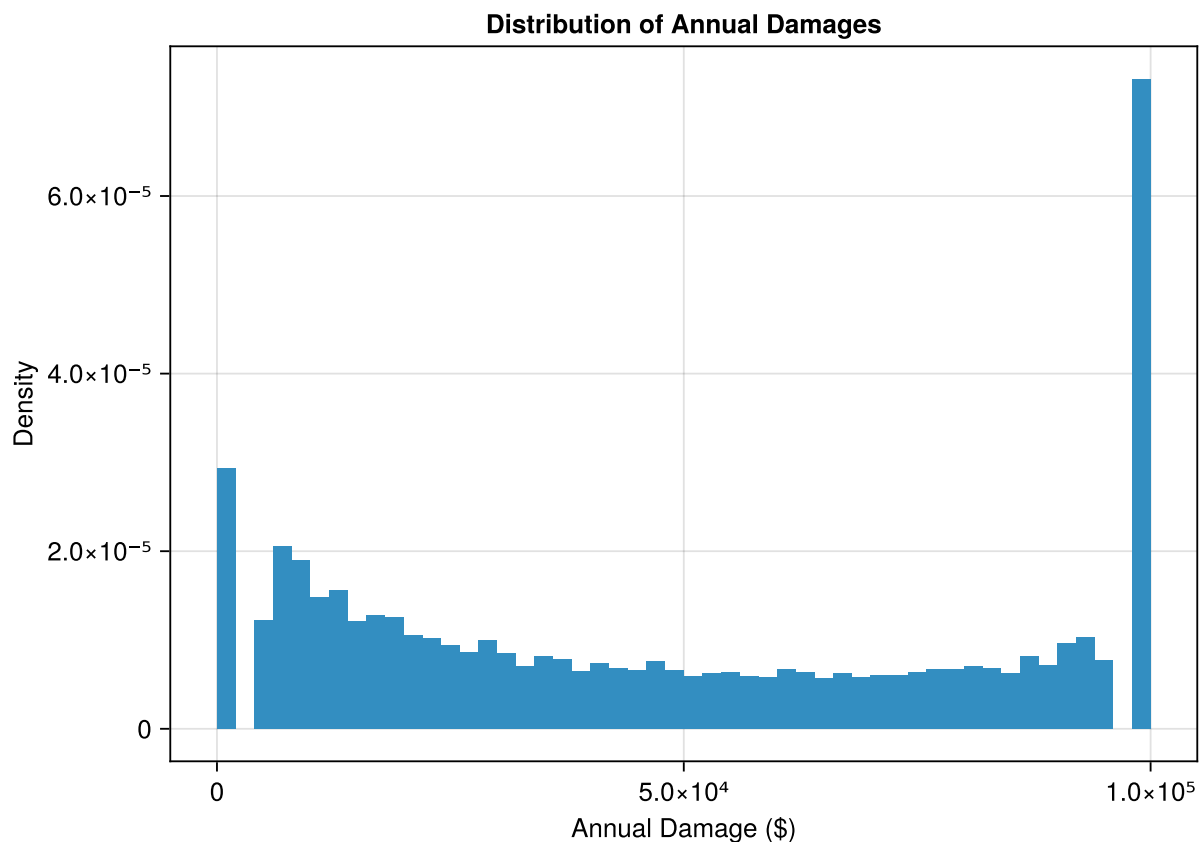


Figure 5: Distribution of annual damages from Monte Carlo convolution, shown as a PDF.

### 3.1.6 Risk Statistics

```
ead_mc = mean(damage_samples)
prob_any_damage = mean(damage_samples .> 0)
prob_severe = mean(damage_samples .> 50_000)

println("Expected Annual Damage (Monte Carlo): \$$ (round(ead_mc; digits=2))")
println("Probability of any damage: $(round(prob_any_damage * 100; digits=1))%")
println("Probability of damage > \$50k: $(round(prob_severe * 100; digits=1))%")
```

#### Line 1

Mean of samples estimates the expected value

#### Line 2

Fraction of samples with positive damage

#### Line 3

Fraction of samples exceeding \$50,000

```
Expected Annual Damage (Monte Carlo): $48917.84
Probability of any damage: 94.1%
Probability of damage > $50k: 45.9%
```

## 3.2 Structuring Analysis with SimOptDecisions

Now let's use the **SimOptDecisions.jl** framework to structure our analysis. This framework helps organize risk analysis by clearly separating:

- **Config:** Fixed parameters (house value, house elevation)
- **Scenario:** One state of the world (one flood event)

```
include("house_elevation.jl")
```

#### Line 1

Load type definitions from a separate file

The `house_elevation.jl` file defines:

- `HouseConfig`: holds house value and base elevation
- `FloodScenario`: represents ONE flood event (one water level)
- `depth_damage()`: logistic depth-damage function
- `compute_damage()`: computes dollar damage for a house given a flood scenario

#### **i** Note

**Key insight:** A `FloodScenario` represents a single state of the world – one sampled water level. It is NOT a probability distribution. We create many scenarios by sampling from the hazard distribution.

### 3.2.1 Visualizing the Depth-Damage Function

```
let
  depths = range(-2, 12; length=200)
  damages = [depth_damage(Float64(d)) for d in depths]
  fig = Figure()
  ax = Axis(
    fig[1, 1];
    xlabel="Flood Depth Above First Floor (ft)",
    ylabel="Damage Ratio",
    title="Depth-Damage Function"
  )
  lines!(ax, depths, damages; linewidth=2)
  vlines!(ax, [0]; color=:gray, linestyle=:dash, label="Floor level")
  fig
end
```

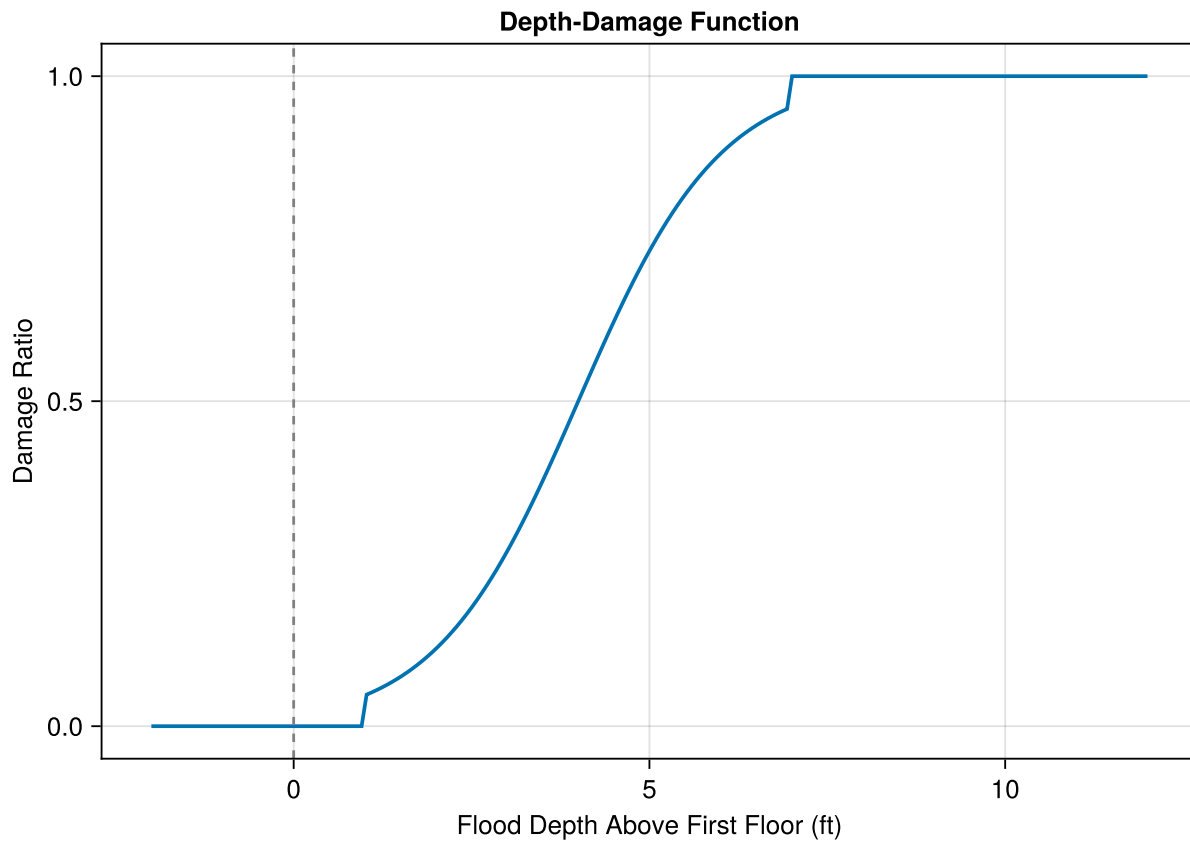


Figure 6: Logistic depth-damage function from SimOptDecisions.

### 3.2.2 Single House Analysis

Let's analyze a house at 4 ft elevation:

```
my_house = HouseConfig(house_value=250_000.0, base_elevation=4.0)
```

**Line 1**

Create a configuration object with keyword arguments

Now we create a FloodScenario for each sampled water level and compute damage:

```
scenarios = [FloodScenario(h) for h in hazard_samples]
house_damages = [compute_damage(my_house, s) for s in scenarios]
```

**Line 1**

Wrap each water level sample in a FloodScenario object

**Line 2**

Compute damage for each scenario using our house configuration

```
let
    fig = Figure()
    ax = Axis(
        fig[1, 1];
        xlabel="Annual Damage (\$)",
        ylabel="Density",
        title="Damage Distribution for House at 4 ft Elevation"
    )
    hist!(ax, house_damages; bins=50, normalization=:pdf)
    fig
end
```

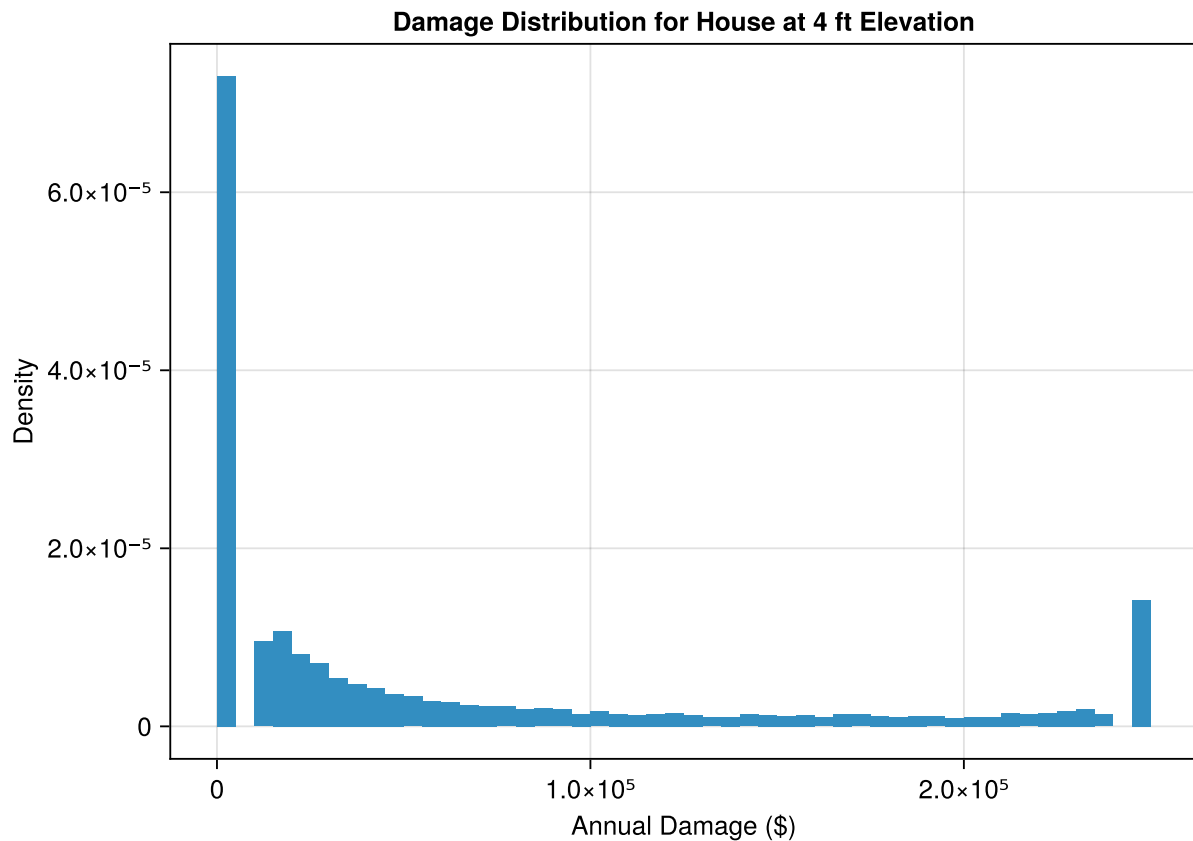


Figure 7: Distribution of annual damages for a single house at 4 ft elevation.

```
ead_house = mean(house_damages)
prob_damage_house = mean(house_damages .> 0)
prob_severe_house = mean(house_damages .> 50_000)

println("Expected Annual Damage: \${round(ead_house; digits=2)}")
println("Probability of any damage: \${round(prob_damage_house * 100; digits=1)}%")
println("Probability of damage > \$50k: \${round(prob_severe_house * 100; digits=1)}%")
```

```
Expected Annual Damage: $63915.29
Probability of any damage: 63.5%
Probability of damage > $50k: 36.7%
```

### 3.3 Community Risk Analysis

Now let's extend our analysis to a community of 100 houses at different elevations. This represents a "city on a wedge" where houses closer to the water are at lower elevations.

#### 3.3.1 Setting Up the Community

```
n_houses = 100
```

```

house_elevations = rand(Uniform(3.0, 12.0), n_houses)
house_value_community = 250_000.0

houses = [HouseConfig(house_value=house_value_community, base_elevation=e) for e in
house_elevations]

```

#### Line 3

Random elevations between 3 and 12 feet

#### Line 4

All houses have the same value for simplicity

#### Line 6

Create a vector of HouseConfig objects

### 3.3.2 Computing EAD for Each House

**Key insight:** All houses experience the **same storm** (same hazard samples). The only difference is their elevation.

```

expected_damages = zeros(n_houses)

for i in 1:n_houses
    damages_i = [compute_damage(houses[i], s) for s in scenarios]
    expected_damages[i] = mean(damages_i)
end

```

#### Line 1

Pre-allocate a vector to store results

#### Line 3

Loop over each house

#### Line 4

Compute damage for all 10,000 scenarios

#### Line 5

Store the mean (expected) damage

```

let
    fig = Figure()
    ax = Axis(
        fig[1, 1];
        xlabel="House Elevation (ft)",
        ylabel="Expected Annual Damage (\$)",
        title="EAD vs. Elevation for 100 Houses"
    )
    scatter!(ax, house_elevations, expected_damages; markersize=8)

```

```
fig
end
```

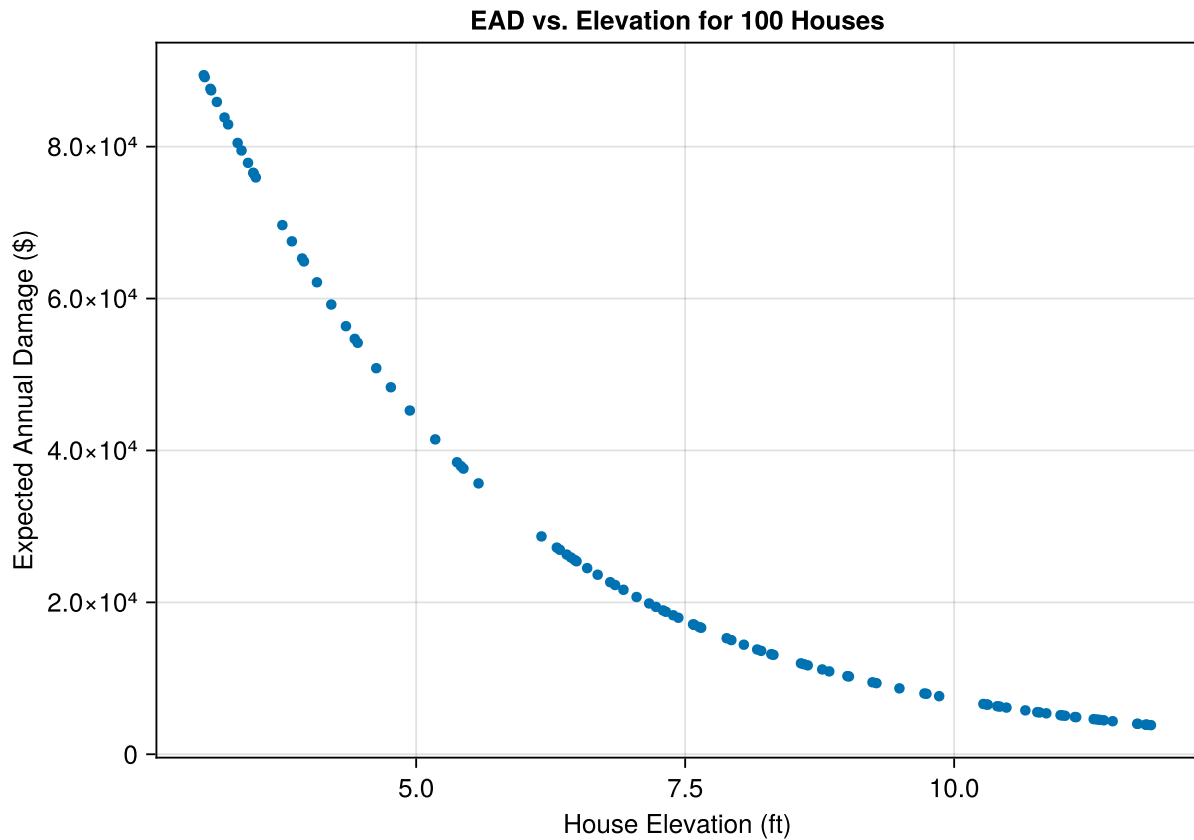


Figure 8: Expected annual damage decreases nonlinearly with house elevation.

#### 💡 Your Response

*Why is the relationship between elevation and expected damage nonlinear? What causes the steep drop-off as elevation increases?*

*Replace this text with your explanation.*

### 3.3.3 Community-Level Damage Distribution

To understand total community risk, we compute total damage for each Monte Carlo sample:

```
total_damages = zeros(n_samples)

for j in 1:n_samples
    scenario_j = scenarios[j]
    total_damages[j] = sum(compute_damage(houses[i], scenario_j) for i in 1:n_houses)
end
```



**Line 1**

One total damage value per scenario (possible year)

**Line 3**

Loop over scenarios (not houses!)

**Line 5**

Sum damage across all 100 houses for this scenario

```
let
  fig = Figure()
  ax = Axis(
    fig[1, 1];
    xlabel="Total Community Damage (\$)",
    ylabel="Density",
    title="Distribution of Annual Community Damages"
  )
  hist!(ax, total_damages; bins=50, normalization=:pdf)
  fig
end
```

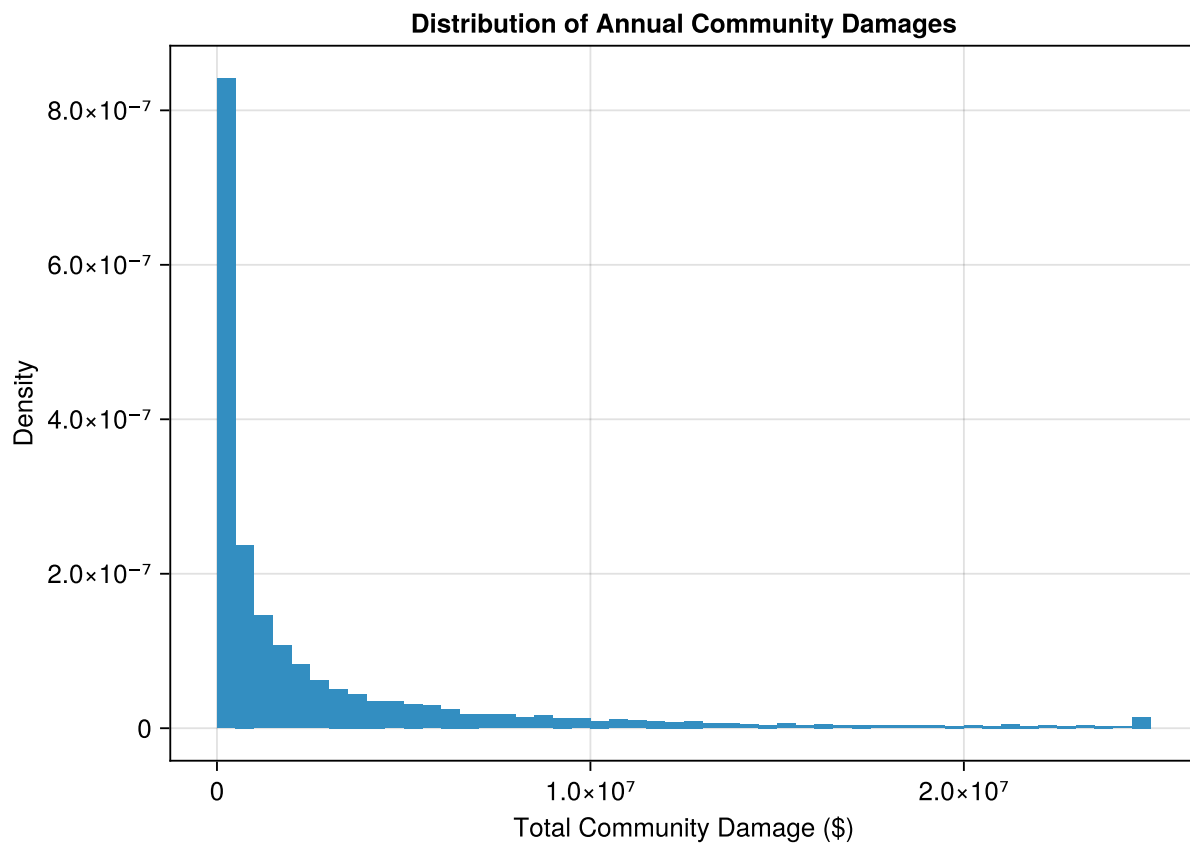


Figure 9: Distribution of total annual damages across the community.

```
total_ead = mean(total_damages)
println("Total Community EAD: \${round(total_ead / 1e6; digits=3)} million")
println("Sum of individual EADs: \${round(sum(expected_damages) / 1e6; digits=3)} million")
```

```
Total Community EAD: $2.841 million
Sum of individual EADs: $2.841 million
```

### 💡 Your Response

*Why are the total community EAD and the sum of individual EADs equal (or nearly equal)? Would they still be equal if we used different hazard samples for each house?*

*Replace this text with your answer.*

## 3.4 The SimOptDecisions Framework

The SimOptDecisions.jl package provides a structured way to think about decisions under uncertainty. Here's the key vocabulary:

| Term            | What it means                      | In this lab                 |
|-----------------|------------------------------------|-----------------------------|
| <b>Config</b>   | Fixed parameters that don't change | House value, base elevation |
| <b>Scenario</b> | One state of the world             | One flood water level       |
| <b>State</b>    | Evolving conditions over time      | (Not used today)            |
| <b>Action</b>   | A decision                         | (Not used today)            |
| <b>Policy</b>   | A decision rule                    | (Not used today)            |

Today we focused on **risk analysis**: understanding the distribution of damages given uncertainty in hazards.

### i Note

In future labs, we'll use SimOptDecisions to:

- Add **State** to track conditions that change over time (like sea level rise)
- Add **Actions** and **Policies** to represent decisions (like how high to elevate)
- Compare different policies across many scenarios

## 3.5 Reflection Questions

Answer these questions in a few sentences each:

### 💡 Question 1

*How does uncertainty in the hazard distribution propagate to uncertainty in damages through convolution?*

*Replace this text with your answer.*

### 💡 Question 2

*Why is expected damage NOT equal to damage at the expected flood stage? (Hint: think about Jensen's inequality and the shape of the damage function.)*

*Replace this text with your answer.*

### 💡 Question 3

*What assumptions did we make that might not hold in reality? List at least three and explain why they matter.*

*Replace this text with your answer.*

## 3.6 Summary

In this lab, you learned to:

1. **Convolve** hazard distributions through damage functions analytically and via Monte Carlo
2. Work with the **GEV distribution** for extreme event modeling
3. Apply **depth-damage functions** to translate hazard into losses
4. Analyze **community-level risk** with 100 houses at varying elevations
5. Structure risk analysis using **SimOptDecisions.jl** (Config + Scenario)

The key insight: **convolution is the mathematical bridge** between “how bad could the flood be?” and “how much will it cost?”

## 3.7 Submission

1. Push your completed notebook to GitHub
2. If the formatter creates a pull request, approve it and re-render
3. Submit the link to your repository on Canvas

## Checklist

Before submitting, make sure you have completed these tasks. Each task involves **modifying existing code**, not writing from scratch.

- ☐ **Setup:** Run all setup cells and verify packages load without error
- ☐ **Hand calculation:** Compute the EAD for the discrete example and write your answer

- **[?] GEV exploration:** Change at least two GEV parameters and describe how they affect the distribution (especially how  $\xi$  affects the right tail)
- **[?] Homeowner question:** Explain what EAD hides about risk
- **[?] SimOptDecisions house:** Change `my_house` elevation from 4.0 to 2.0 and observe how the damage distribution changes
- **[?] EAD vs elevation:** Answer why the relationship is nonlinear
- **[?] Community EAD:** Answer why total EAD equals sum of individual EADs
- **[?] Reflection questions:** Answer all three reflection questions
- **[?] LLM exploration:** Use GitHub Copilot, Claude Code, or another AI assistant to explore this codebase. Ask it to explain why a specific piece of code works the way it does. In your notebook, briefly describe one thing you learned about the code from the LLM.
- **[?] Render:** Re-render the notebook to make sure all outputs are up to date

### 3.8 References

## Bibliography